

The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of how vaccines transformed public health from 1796 through 2020 — and where coverage gaps remain.

1796
first vaccine

1980
smallpox eradicated

2020
COVID-19 licensed <1 year

Prepared for public health professionals and policy analysts

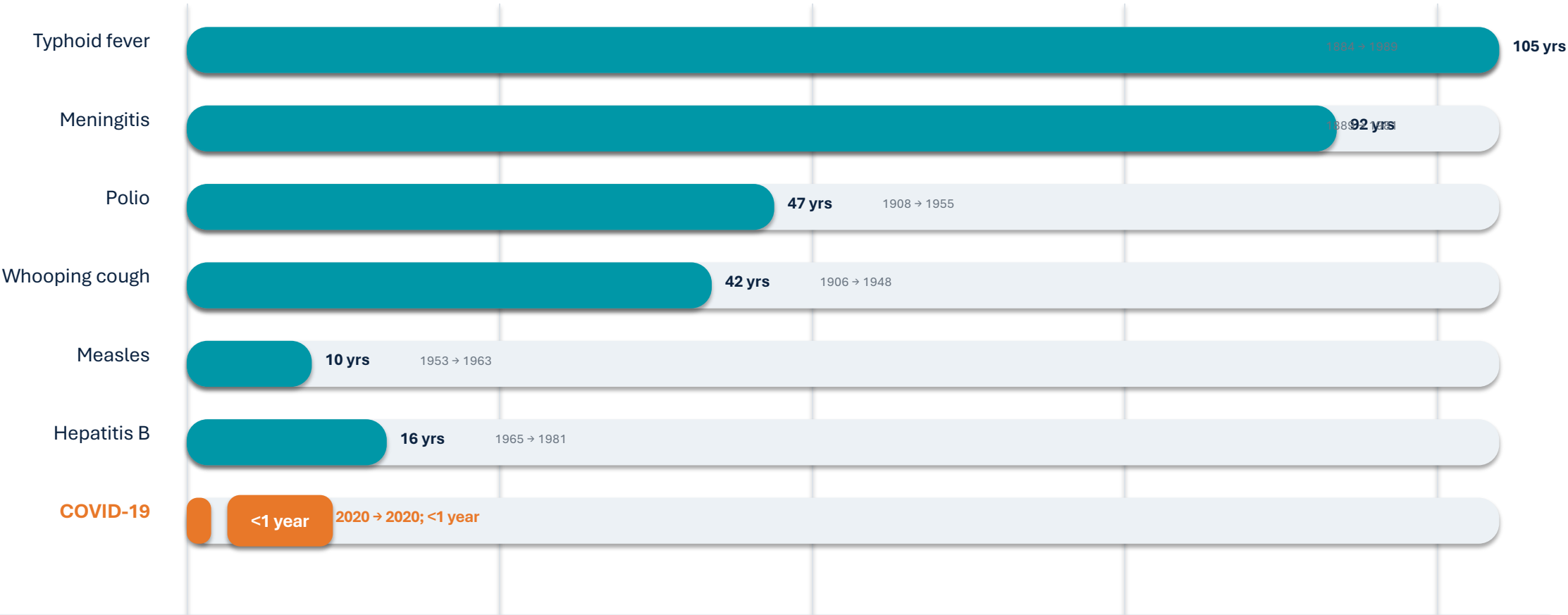
Source: Our World in Data — Vaccination

Development speed,
disease reduction,
and coverage gaps

- DATA-FORWARD
- AUTHORITATIVE
- ACCESSIBLE

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure has compressed from more than a century to under a year.



Modern tools — **advanced microscopy, genome sequencing, mRNA technology, and virus-like particles** — enabled faster and more precise vaccine development.

Source: Our World in Data vaccination brief; timeline values from pathogen identification to vaccine licensure.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Roush and Murphy data show elimination for several diseases while pneumococcal disease and hepatitis retain measurable burden.

Disease	Pre-vaccine cases / million/yr	Post-vaccine cases / million/yr	Case reduction	Pre-vaccine deaths / million/yr	Death reduction
Diphtheria	158	0	✓ 100%	13.7	100%
Smallpox	250	0	✓ 100%	2.9	100%
Acute polio	141	0	✓ 100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

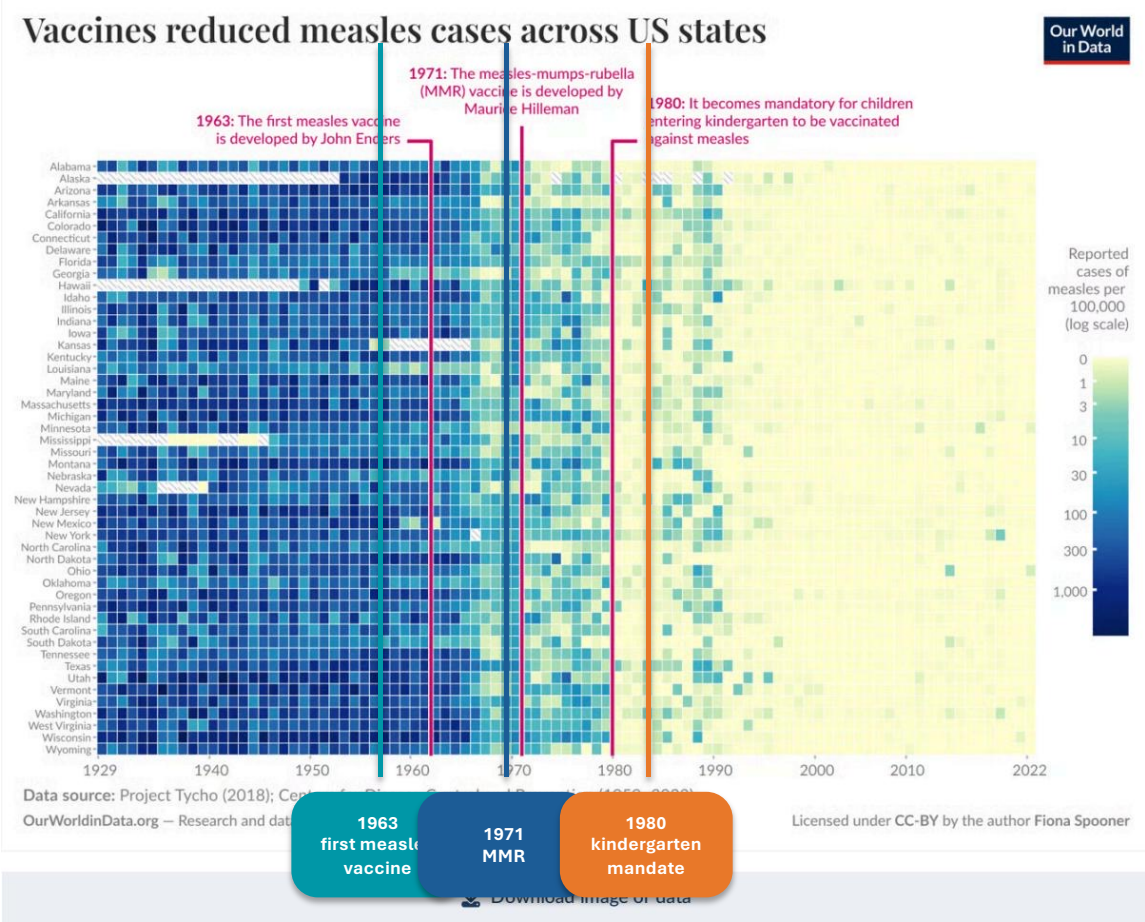
100% case + death reduction

Diphtheria, smallpox, and acute polio reached zero post-vaccine cases per million per year in the US.

Source: Our World in Data vaccination brief; Roush and Murphy (2007) where noted.

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

The heatmap tracks measles cases per 100,000 across states from 1929–2022, with milestone years reshaping incidence.



Before 1963

Nearly every state experienced high incidence: 100–1,000+

After 1980

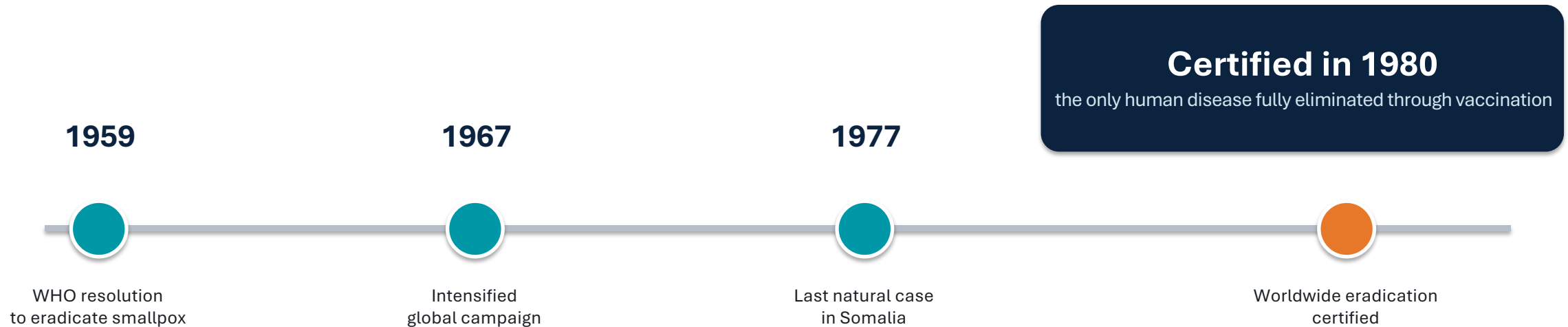
Cases dropped to near zero across all states, with only sporadic outbreaks

Policy signal

Mandates and combination vaccines turned high incidence into near zero

Smallpox: The Only Human Disease Eradicated Through Vaccination

Smallpox eradication required global coordination, a feasible biological target, and a cheap, effective vaccine.



Why smallpox was uniquely suited for eradication

Humans only

The virus infected only humans.

Visible symptoms

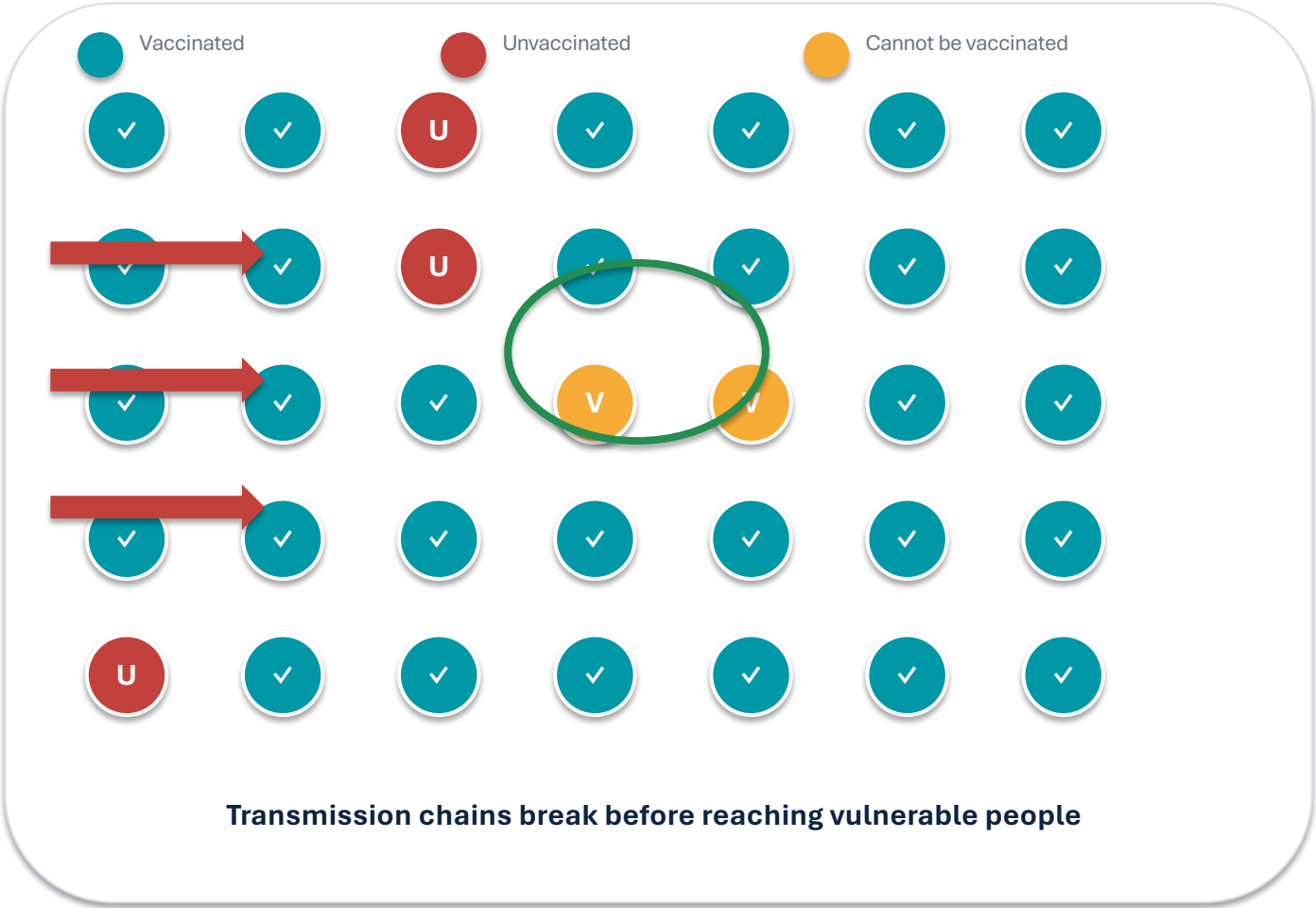
Symptoms were distinguishable.

Practical vaccine

The vaccine was cheap and effective.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When enough people are vaccinated, pathogens cannot spread effectively — protecting people who cannot be vaccinated.



Population-level effect

Vaccines protect individuals and reduce community spread.

Who is shielded?

Newborns, immunocompromised individuals, and cancer patients on

Policy implication

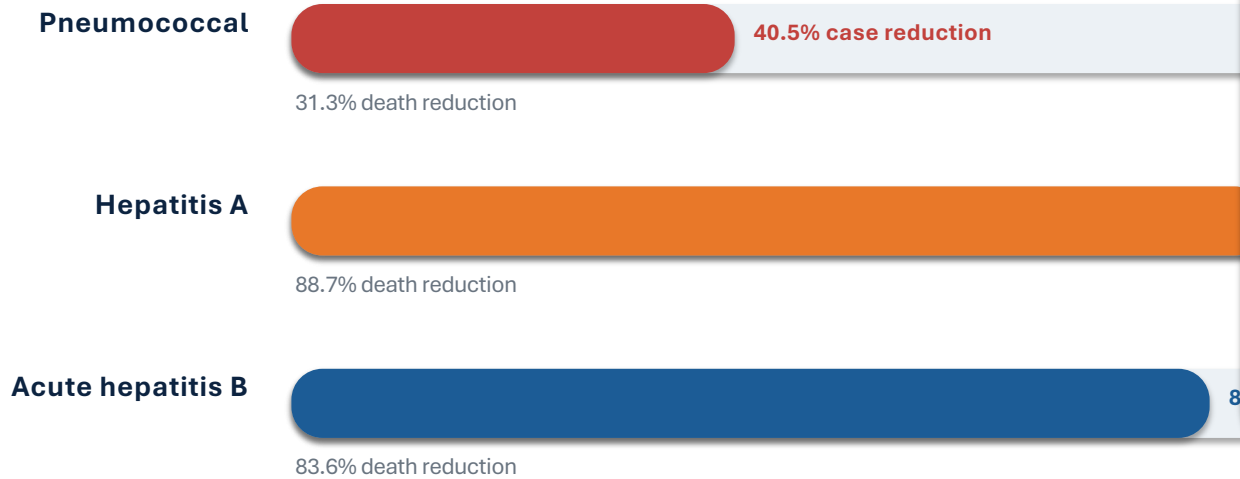
Coverage is infrastructure: gaps become transmission pathways.

Source: Our World in Data vaccination brief; herd immunity description from source brief.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Incomplete disease control points to the operational challenge: access, supply, trust, and continued vaccine improvement.

Case reduction vs. remaining burden



Persistent barriers

- Not everyone has access to vaccines globally
- Supply shortages put children at risk
- Poor scientific understanding can reduce trust and uptake

The next frontier is not only discovery; it is delivery: **improve access, coverage, trust, and newer vaccines.**

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

A policy agenda for sustaining the gains of vaccination and closing the remaining gaps.

Elimination is possible

100% elimination of cases and deaths for diphtheria, smallpox, and acute polio in the US.

Development has accelerated

Timelines collapsed from over 100 years for typhoid to under 1 year for COVID-19.

Measles control is policy-sensitive

Cases dropped to near zero across all 50 states after the 1980 kindergarten vaccination mandate.

Smallpox proves global eradication

Certified in 1980 after a global campaign; still the only human disease eradicated through vaccination.

Coverage gaps remain

Pneumococcal disease shows 40.5% case reduction; hepatitis A and B remain incomplete.

Call to action for global health systems

Strengthen access, stabilize supply, build scientific understanding, and sustain high coverage so community protection is sustained